

Team 20: Generation for Recommendation

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Milestone 3

1. Methodology

In Milestone 3, we kept the main end-to-end pipeline from Milestone 2: embedding-based user profiling, LLM-based prompt construction, Suno API music generation, CLAP-based reranking, and Streamlit visualization. The main new contribution in this milestone is replacing the base CLAP setting with fine-tuned CLAP and adding a more structured evaluation framework.

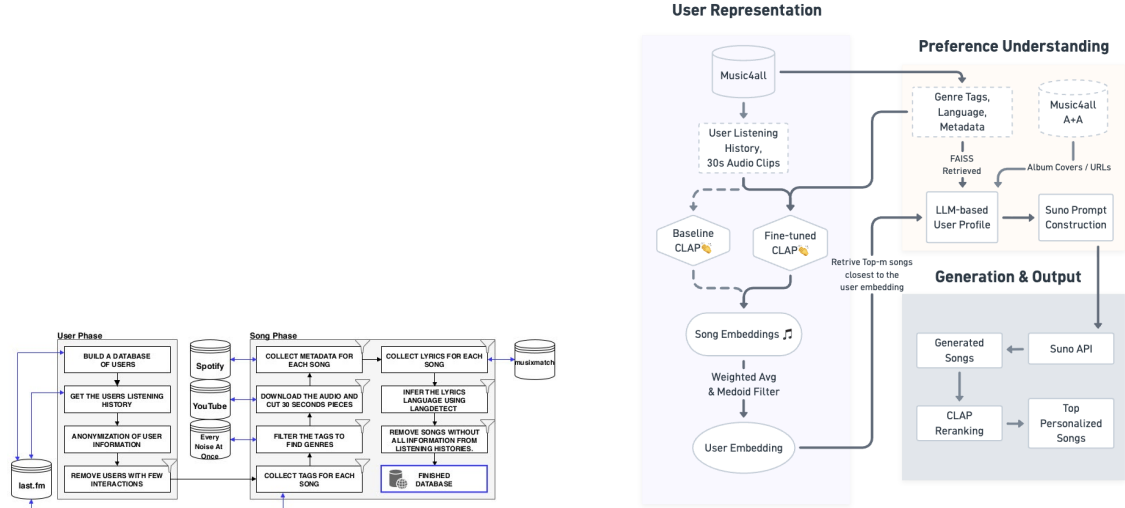


Figure 1: End-to-end overview of the Gen4Rec system. (a) Music4All preprocessing and metadata construction pipeline. (b) Personalized music generation workflow using fine-tuned CLAP embeddings, FAISS retrieval, LLM-based profile construction, Suno generation, and CLAP reranking.

For fine-tuned CLAP, we trained the model on Music4All audio paired with genre and tag-based text descriptions, so that the embedding space better captures music-domain similarity. Each song was split into multiple audio chunks, pooled into one track-level

representation, and optimized with an audio-text contrastive objective. For audio-to-text matching, we used a soft target formulation:

$$y = (1 - \lambda)I + \lambda \cdot \text{softmax}(\tau \cdot S_{\text{text}})$$

where I is the one-hot matching target, S_{text} is the text-text similarity matrix, λ controls how much semantic similarity is mixed into the target, and τ controls the sharpness of the softmax distribution. This allows songs with similar genre or tag descriptions to be treated as partially related instead of completely negative pairs, making cosine similarity in the fine-tuned CLAP space more meaningful for personalization.

We also designed two evaluation phases. Phase 1 compares base CLAP and fine-tuned CLAP in a blind retrieval study. Given the same text prompts, both models retrieve music clips, and users judge which results better match the prompt. Phase 2 evaluates the full personalized generation pipeline. Each participant provides a custom song list, which is converted into 30-second clips and embedded with fine-tuned CLAP to form a synthetic user embedding. This embedding is then used for retrieval, profile construction, LLM prompt generation, Suno generation, and CLAP-based reranking.

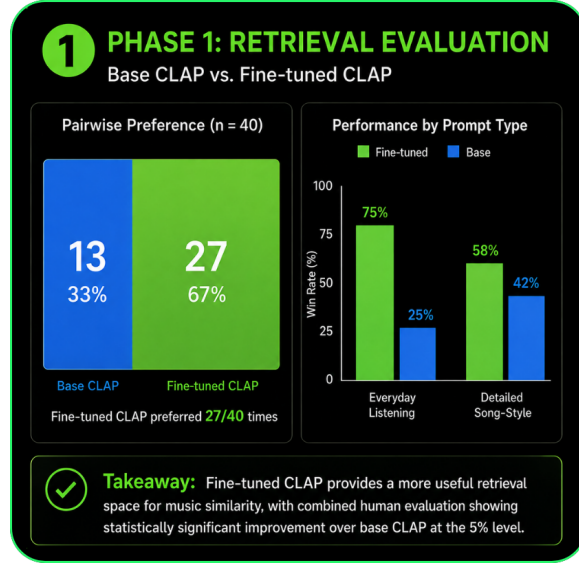
Overall, Milestone 3 makes Gen4Rec more evaluation-driven while preserving the core pipeline built in Milestone 2. We also added the fine-tuned CLAP component to improve music-domain similarity matching, and built a Streamlit demo to visualize the end-to-end workflow, including user profiling, prompt generation, generated outputs, reranking scores, and evaluation results. Full implementation details are available in our project repository: <https://github.com/J-hjr/Gen4Rec/blob/main/implementation.md>

2. Results

In Milestone 3, we evaluated Gen4Rec through two phases: a blind retrieval evaluation and a personalized generation evaluation. The main goal was to test whether replacing base CLAP with fine-tuned CLAP improves music retrieval and whether the full pipeline can generate personalized music recommendations from user listening histories.

For **Phase 1**, we conducted a blind retrieval study comparing base CLAP and fine-tuned CLAP. Detailed evaluation metrics are provided in **Appendix 2**, and the full prompt set used in the retrieval evaluation is included in **Appendix 4**. We used 20 evaluation prompts in total. The first 8 prompts focused on everyday listening scenarios, such as rainy nights, morning coffee, or workout energy. The remaining 12 prompts were more detailed song-style prompts, summarized from real users’ listening histories. For each prompt, the base CLAP model retrieved 2 clips, and the fine-tuned CLAP model retrieved 2 clips. The four clips were shuffled and presented anonymously to evaluators, so they did not know which model produced each result. For each prompt, participants ranked the four clips from best match to worst match. We assigned scores based on the ranking position: best(5), second(3), third(1), worst(0). In total is 8.

The evaluation was completed by 4 evaluators, who divided the prompts among themselves. To reduce evaluation bias, the full set of 20 prompts was evaluated twice by



them (different person done different part, no repeated evaluation for the same prompt done by one person), and the final results were averaged across the two evaluation tables. Across all prompts, fine-tuned CLAP was selected 27 out of 40 times, suggesting that fine-tuned CLAP generally outperformed base CLAP in blind retrieval. This result indicates that fine-tuning CLAP on Music4All genre and tag descriptions improves the embedding space for music-domain retrieval.

However, the improvement was not uniform across prompt types. Fine-tuned CLAP performed better on the everyday scenario prompts than on the more detailed, song-focused prompts. Across the two evaluation tables, the average fine-tuned CLAP ranking score was 5.94/8 for prompts p01–p08, compared with 5.08/8 for prompts p09–p20. This suggests that the fine-tuned model is stronger when prompts describe broad moods, contexts, and listening scenarios, while its advantage becomes less stable when prompts contain more specific or complex musical style descriptions.

To further test whether this improvement was statistically meaningful, we conducted a one-sided sign test on the combined prompt-level results. The null hypothesis was $H_0 : p = 0.5$, meaning that fine-tuned CLAP and base CLAP are equally likely to win on a prompt. The alternative hypothesis was $H_1 : p > 0.5$, where p represents the probability that fine-tuned CLAP outperforms base CLAP on a prompt.

Across the two evaluation rounds, fine-tuned CLAP won 27 out of 40 prompt-level comparisons, while base CLAP won 13. This gives a one-sided p-value of 0.0192. Since $0.0192 < 0.05$, we reject H_0 at the 5% significance level. This provides stronger evidence that fine-tuned CLAP performs better than base CLAP in human retrieval evaluation.

For **Phase 2**, we evaluated the full personalized generation pipeline with 6 participants. Each participant provided a custom list of 10 songs, which were converted into 30-second audio clips and embedded using fine-tuned CLAP to form a synthetic user embedding. This embedding was then used for Music4All retrieval, user profile construction,

LLM-based prompt generation, Suno music generation, and CLAP-based reranking. For each participant, we generated 6 Suno candidates using 3 generation calls, with each call returning 2 songs. We then used fine-tuned CLAP cosine similarity to rerank the 6 candidates and selected the top 2 songs for user listening and feedback. Detailed evaluation metrics for the personalized generation study are provided in **Appendix 3**. Participants were fully blinded when listening to the 2 songs, only after they filled out the evaluation form can they view the cosine score for each song.

The Phase 2 results suggest that the personalized pipeline can produce outputs that are generally aligned with users’ listening preferences. Across the 6 participants, the generated user profiles received accuracy scores ranging from 7/10 to 10/10, with an average score of approximately 8.5/10. This indicates that the retrieval and profile construction stages were usually able to capture users’ broad genre preferences, mood patterns, vocal or instrumental tendencies, and production-style preferences.

The CLAP-based rerank results were also mostly consistent with human preference. In 5 out of 6 cases, the top-ranked song selected by CLAP was also the user’s preferred song. The only mismatch occurred in Irene’s case, where CLAP ranked Track B slightly higher, but the user preferred Track A because it sounded clearer and more coherent. This suggests that CLAP cosine similarity is a useful rerank signal, but it cannot fully replace human judgment on subjective qualities such as melody clarity, emotional coherence, and overall listening experience.

Participant	Profile Score	Top CLAP Score	CLAP = User Favorite	Top Song Rating
Eason	8/10	0.2200	Yes	5/5
Irene	8/10	0.6090	No	4/5
Tony	10/10	0.8236	Yes	5/5
Nancy	7/10	0.6369	Yes	5/5
Jerry	9/10	0.7886	Yes	5/5
Ryan	9/10	0.6289	Yes	5/5

Table 1: Phase 2 personalized generation evaluation summary. “Top Song Rating” refers to the user’s preference match score for their preferred generated song.

Human feedback further supports the effectiveness of the personalized generation pipeline. Five of the six users gave their preferred generated song a 5/5 preference match score, suggesting that the system was usually able to generate music that matched the user’s taste. One participant Irene’s preferred song received a 4/5 preference match score, and her feedback showed that the output was directionally aligned with her upbeat and rhythmic preference, but the melody and arrangement felt somewhat cluttered. This case highlights that even when the embedding score is high, subjective musical coherence still matters.

Overall, Phase 2 shows that Gen4Rec can complete the full personalization workflow on real participant inputs and produce recommendation candidates that are often preferred by users. The results also suggest that fine-tuned CLAP is useful not only for constructing user embeddings, but also for reranking generated songs. However, because the evaluation only includes 6 participants and the final judgment depends on subjective listening feed-

back, these results should be interpreted as preliminary evidence rather than a definitive conclusion.

3. Analysis

The **Phase 1** results suggest that fine-tuned CLAP provides a more useful embedding space for music retrieval than the base CLAP model. Since the fine-tuned model was selected 27/40 times, it shows a clear but not overwhelming improvement over the base model. This supports our decision to replace the base CLAP setting with fine-tuned CLAP in Milestone 3. The improvement is meaningful because our recommendation pipeline depends heavily on cosine similarity in the CLAP embedding space. If the embedding space better captures music-domain similarity, then user profiling, retrieval, generation prompt construction, and reranking can all become more reliable.

At the same time, the results also show that fine-tuning does not solve all retrieval problems. The fine-tuned model performed better on everyday scenario prompts than on detailed song-style prompts. One possible explanation is that everyday prompts are more aligned with genre, mood, and high-level tag information, which are exactly the types of signals used during fine-tuning. In contrast, the later prompts contain more specific musical descriptions summarized from real listening histories. These prompts may involve more subtle combinations of instrumentation, production style, emotional tone, and artist-level references, which are harder for CLAP to capture through genre/tag supervision alone. This finding is important because it shows both the strength and limitation of our fine-tuning strategy. The semantic soft target design helps the model learn softer relationships between related genres and tags, but it may still be too coarse for highly detailed music preference modeling.

The **Phase 2** results suggest that the full personalized generation pipeline is technically feasible and can produce music that users perceive as aligned with their tastes. Across the 6 participants, profile accuracy scores ranged from 7/10 to 10/10, with an average of approximately 8.5/10. This indicates that the retrieval and profile construction stages can usually capture meaningful preference signals from only 10 input songs, including genre, mood, vocal style, tempo, and production texture.

The CLAP-based reranking results also provide support for using fine-tuned CLAP as an automatic selection mechanism. In 5 out of 6 cases, the top-ranked song selected by CLAP was also the user’s preferred song. This suggests that cosine similarity in the fine-tuned CLAP space is a useful signal for choosing among multiple generated candidates. Since Suno generation is stochastic, reranking is important because different calls using the same prompt can still produce songs with different structures, moods, and sound quality.

At the same time, the results show that CLAP similarity cannot fully replace human judgment. Irene’s case is the clearest limitation: although CLAP assigned a slightly higher score to Track B, the user preferred Track A because it sounded clearer and more coherent. This suggests that embedding similarity can capture broad preference alignment, but it may miss subjective musical factors such as melody clarity, arrangement structure, emotional smoothness, and whether the song feels too cluttered.

Another important finding is that profile accuracy and final recommendation quality are related but not identical. For example, Nancy’s profile received a lower score because it overemphasized darker emo-rap and experimental hip-hop elements while missing her softer and more romantic atmosphere. However, her final preferred generated song still received a 5/5 preference match score, and CLAP reranking correctly selected it as the top candidate. This suggests that even when the profile is imperfect, the generation and reranking stages may still recover a strong personalized output.

Overall, Phase 2 supports the main idea of Milestone 3: fine-tuned CLAP can be used not only for user embedding construction, but also for selecting generated songs that are likely to match user preferences. However, the conclusion should remain cautious because the evaluation includes only 6 participants and relies on subjective listening feedback. A stronger future evaluation would compare personalized prompts against generic or random baseline prompts across more users. Still, the current results provide preliminary evidence that Gen4Rec can generate personalized music recommendations that are often preference-aligned and interpretable.

4. Conclusions

Overall, Gen4Rec shows that a generative recommendation pipeline can move beyond retrieving existing songs and instead create new music candidates personalized to a user’s listening history. A key strength of the system is its end-to-end design: it connects user embedding construction, LLM-based listener profiling, Suno-based music generation, CLAP-based reranking, and evaluation in a single workflow. Our evaluation outputs suggest that the fine-tuned CLAP model performs better than the base CLAP setting, especially because it better captures music-domain similarity from genre and tag information. In the generation stage, the Suno outputs also generally fit the audience’s listening taste, and the CLAP-based reranking further helps select the generated songs that are most aligned with the user embedding and reference tracks.

Another strength is interpretability. The system does not only output generated songs; it also provides user profiles, representative genres, tags, artists, tracks, prompts, reranking scores, and evaluation metrics. This makes the recommendation and generation process easier to inspect and debug compared with a fully black-box music generation system.

However, the system still has several limitations. First, retrieval currently depends only on the Music4All dataset, so the recommendation space is constrained by the songs, genres, and metadata available in that dataset. If certain genres, languages, or niche styles are underrepresented, the generated profiles and prompts may inherit those biases. Second, the current pipeline mainly relies on English-language metadata and lyrics, which limits its ability to represent multilingual listening preferences. Third, the genre and tag space is limited, so some users’ musical taste may be compressed into broad labels rather than captured in full detail. Finally, CLAP similarity is only a proxy for human preference. A generated song may score highly in embedding space but still feel less appealing to a listener because of vocals, lyrics, structure, production quality, or emotional fit.

For future work, we plan to expand the retrieval database beyond Music4All, include more diverse genres and non-English music, and improve the metadata used for profile construction. We also want to collect more human evaluation results to compare automatic CLAP-based rankings with real listener preferences. Another direction is to improve reranking by combining CLAP similarity with additional signals such as lyrical relevance, diversity, novelty, and direct user feedback. In the longer term, Gen4Rec could become a more robust personalized music generation system by continuously learning from user ratings and updating the user profile over time.

5. Work Flow

5.1 Timeline of Tasks

- **Week 11:** Built the evaluation user-study pipeline and scripts. Implemented Phase 1 for the base CLAP vs. fine-tuned CLAP retrieval comparison, and Phase 2 for personalized generation evaluation using participant song lists.
- **Week 12:** Ran the user study, collected survey responses, and organized the results. Analyzed whether fine-tuned CLAP improves retrieval quality and whether Suno can generate songs that better match the audience’s taste based on customized playlists.
- **Week 13:** Finalized the Streamlit demo and polished the end-to-end pipeline. Refined the project Markdown files, including the README, implementation documentation, and Streamlit usage guide. Completed Milestone 3 and prepared the presentation.

5.2 Each Member’s Contributions

- **Tony:** Built the scripts for Phase 1 and Phase 2 evaluation, including the base vs. fine-tuned CLAP retrieval comparison and the personalized generation evaluation pipeline. He also adjusted the Streamlit demo.
- **Conny:** Conducted the user survey, collected listener feedback, and organized the evaluation results to compare human preferences with model-based rankings.
- **Jerry:** Added additional dataset support and album-cover enrichment, debugged generation, and helped structure the user-study workflow and demo presentation.

Appendix A.

Metric	Description
Ranking Score	Human ranking score assigned to retrieved clips in the blind retrieval study. The scoring rule was: best = 5, second = 3, third = 1, worst = 0.
Prompt-level Win	Whether the fine-tuned CLAP model achieved a higher total ranking score than the base CLAP model for a prompt.
Sign Test	One-sided sign test comparing prompt-level wins between fine-tuned CLAP and base CLAP retrieval performance.

Table 2: Phase 1 retrieval evaluation metrics used in the blind retrieval study.

Metric	Description	Scale
Profile Accuracy	How well the generated user profile matched the participant’s listening preferences and music taste.	1–10
Preference Match	How well the generated song matched the participant’s personal music preference.	1–5
Sound Quality	Perceived audio quality, coherence, and overall listening experience of the generated song.	1–5
Creativity	Perceived originality and musical creativity of the generated output.	1–5
CLAP-User Agreement	Whether the top-ranked song selected by CLAP reranking matched the participant’s preferred song.	Yes/No

Table 3: Phase 2 personalized music generation evaluation metrics.

Prompt ID	Description
p01	Slow emotional music for walking alone on a rainy night, with gentle piano, quiet, lonely, and reflective atmosphere.
p02	Relaxing acoustic guitar music for drinking coffee in the morning, warm, calm, and comfortable.
p03	Fast high-energy music for working out, powerful, motivating, and intense.
p04	Laid-back summer music for watching the beach sunset, relaxed, sunny, and peaceful.
p05	Cool pop or R&B music for driving through the city at night, stylish, smooth, and confident.
p06	Fun dance music for a small party with friends, cheerful, catchy, and easy to move to.
p07	Calm background music for studying, steady, relaxing, and not distracting.
p08	Smooth romantic dinner music with soft saxophone, warm, intimate, relaxing, and elegant atmosphere.
p09	Cinematic Chinese fantasy music with elegant strings, flute, and mysterious moonlit atmosphere.
p10	Dramatic orchestral battle music with rising tension, heavy percussion, and heroic fantasy energy.
p11	Playful villain-themed music with quirky rhythm, theatrical humor, and festive celebration mood.
p12	Emotional piano and strings music with a gentle, bittersweet romantic atmosphere.
p13	Fast-paced suspense music with chaotic footsteps, nervous rhythm, and chase-like movement.
p14	Ancient ceremonial music with grand drums, bells, and ritual-like atmosphere.
p15	Dreamy mirror-world music with ethereal vocals, soft synths, and surreal fantasy mood.
p16	Epic adventure soundtrack with sweeping strings, strong drums, and a sense of destiny.
p17	Elegant night-time ballad with moonlight, soft melody, and sentimental emotion.
p18	Dark fantasy music with mysterious ambience, low strings, and magical tension.
p19	Lighthearted festive music for a fantasy New Year dinner scene, humorous but warm.
p20	Flowing cinematic instrumental music with wind, clouds, and emotional build-up.

Table 4: Prompt set used in the Phase 1 blind retrieval evaluation comparing baseline and fine-tuned CLAP retrieval performance.